

ICS1512 – Machine Learning Algorithms Laboratory

Experiment 8 Report

Experiment No.	08
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering
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1 Aim and Objectives

To implement and analyse the performance of three unsupervised clustering algorithms on the UCI Human Activity Recognition (HAR) Using Smartphones dataset:

- **Model A** – K-Means clustering, with the number of clusters selected using the elbow method and the Silhouette Score.
- **Model B** – DBSCAN (Density-Based Spatial Clustering of Applications with Noise), with systematic tuning of `eps` and `min_samples`.
- **Model C** – Hierarchical Agglomerative Clustering (HAC), comparing single, complete, average and Ward linkage.

The clusters formed by each algorithm are visualised in two dimensions (PCA and t-SNE), evaluated using internal metrics (Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index) and external metrics against the six true activity labels (Adjusted Rand Index, Normalized Mutual Information), and compared to determine which algorithm and configuration produces the most meaningful clustering of human activities.

2 Problem Statement

Input: 10,299 windows of smartphone accelerometer/gyroscope signal, each pre-processed by the dataset authors into a 561-dimensional feature vector (time- and frequency-domain statistics), together with the (unused-for-training) true activity label and subject id for each window.

Output: A partition of the 10,299 windows into clusters produced independently by K-Means, DBSCAN and hierarchical clustering, with no activity label supplied to any algorithm.

Objective: Determine whether, and how well, unsupervised clustering on the raw sensor-derived features recovers the six known human activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING), and compare the three algorithms using internal and external validity criteria.

3 Dataset Description

Dataset Name	UCI Human Activity Recognition Using Smartphones
Dataset Source	UCI Machine Learning Repository; provided in the course repository at <code>datasets/UCI HAR Dataset/</code>
Number of Samples	10,299 (7,352 train + 2,947 test, combined for this unsupervised experiment)
Number of Features	561 (time- and frequency-domain features, values pre-normalised to $[-1, 1]$ by the dataset authors)
Number of Classes (used only for validation)	6 – WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING
Subjects	30 volunteers, aged 19–48

Missing Values	0 (verified programmatically)
Duplicate Rows	0 (verified programmatically)
Class Balance	Largest/smallest class ratio = 1.38 (Table 2) – reasonably balanced

Table 1: Dataset description

Table 2: Activity distribution (Notebook §5.1)

Activity	Count	Percent	Train	Test
WALKING	1722	16.72%	1226	496
WALKING_UPSTAIRS	1544	14.99%	1073	471
WALKING_DOWNSTAIRS	1406	13.65%	986	420
SITTING	1777	17.25%	1286	491
STANDING	1906	18.51%	1374	532
LAYING	1944	18.88%	1407	537

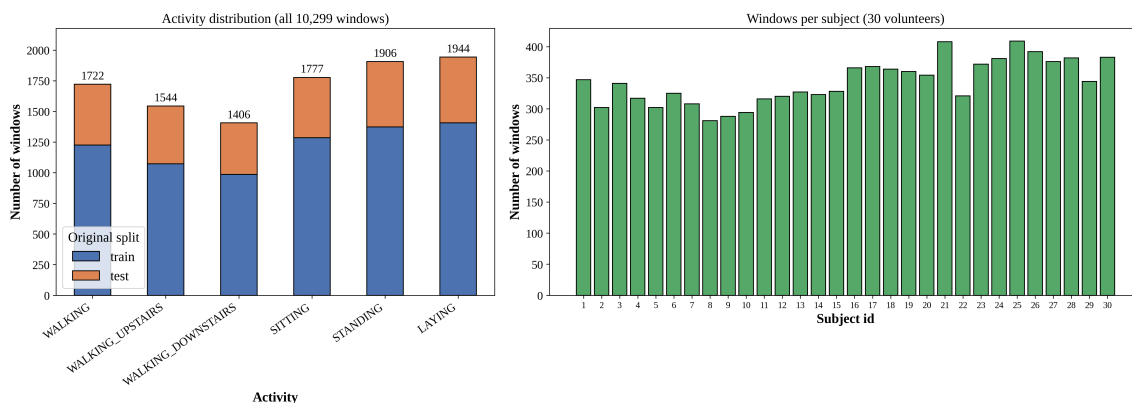


Figure 1: Left: activity distribution across the combined dataset (stacked by original train/test split). Right: number of windows recorded per subject.

Interpretation: All six activities are represented with roughly comparable frequency (13.6%–18.9% of the data each), so no class dominates the unsupervised clustering by sheer volume. Each of the 30 subjects contributed a broadly similar number of windows, so no single subject’s motion pattern can dominate a cluster either.

4 Data Preprocessing

Loading. `X_train.txt`, `X_test.txt`, `y_train.txt`, `y_test.txt`, `subject_train.txt`, `subject_test.txt`, `activity_labels.txt` and `features.txt` were all loaded from the verified dataset path. Column names were taken from `features.txt`; 126 of the 561 names were not unique (repeated across frequency-band sub-features) and were disambiguated by appending the original feature index, so no column was dropped. Train and test partitions were concatenated into a single 10,299-row table, since the original train/test split is a supervised-learning artefact not required for unsupervised clustering.

Quality checks. Dimensions, dtypes, missing values and duplicate rows were checked programmatically: all 561 features are `float64`, with **0 missing values** and **0 duplicate rows**. All feature values lie within $[-1.0, 1.0]$, consistent with the dataset documentation.

Encoding. The six activity labels were mapped from their numeric ids (1–6, from `activity_labels.txt`) to integer codes 0–5 (`y_true`), used exclusively for the external metrics (ARI, NMI) and for colouring the ground-truth reference plots – never as clustering input.

Standardisation. Before scaling, per-feature standard deviations ranged from 0.037 to 0.749, a $20.2\times$ spread. Because K-Means, DBSCAN and (except single/complete which use min/max, still distance-based) hierarchical clustering all rely on Euclidean distance, features with larger native variance would otherwise dominate the distance calculation irrespective of their actual discriminative value. `StandardScaler` was applied to all 561 features, giving mean ≈ 0 (max $|\text{mean}| = 2.6 \times 10^{-15}$) and standard deviation exactly 1 for every feature. All downstream distance-based computations use this standardised matrix.

5 Exploratory Data Analysis

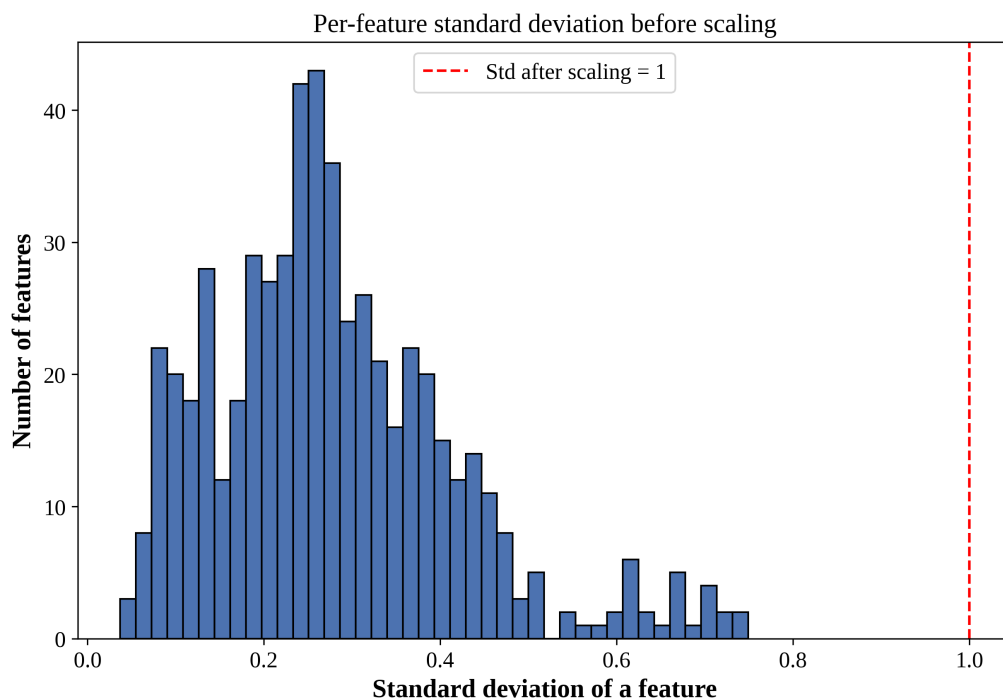


Figure 2: Distribution of per-feature standard deviation before scaling.

Interpretation: Most features have a standard deviation well below 1 before scaling (mean 0.279), with a long right tail; the dashed line at 1 marks where every feature sits after `StandardScaler` is applied, illustrating directly why standardisation changes the geometry of the distance computation.

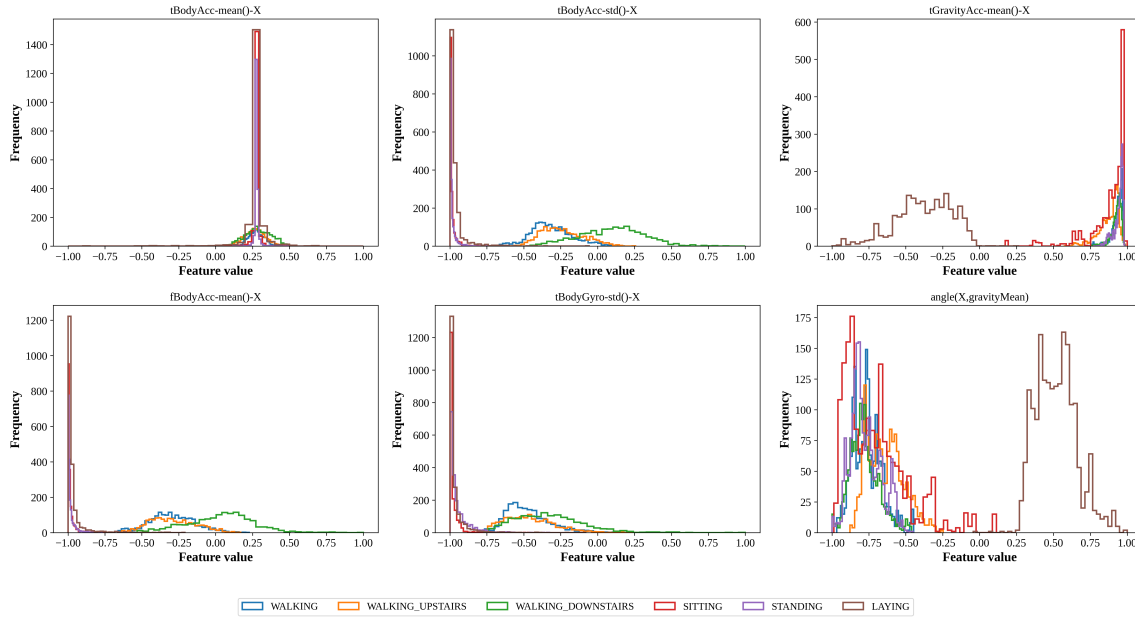


Figure 3: Distribution of six representative features (colour = true activity, for EDA only), split by activity.

Interpretation: Several features show clear separation by activity even individually – e.g. `tGravityAcc-mean()-X` separates the static activities (SITTING/STANDING/LAYING) from the dynamic ones, foreshadowing the strong static-vs-dynamic split found later in PCA and in every clustering algorithm.

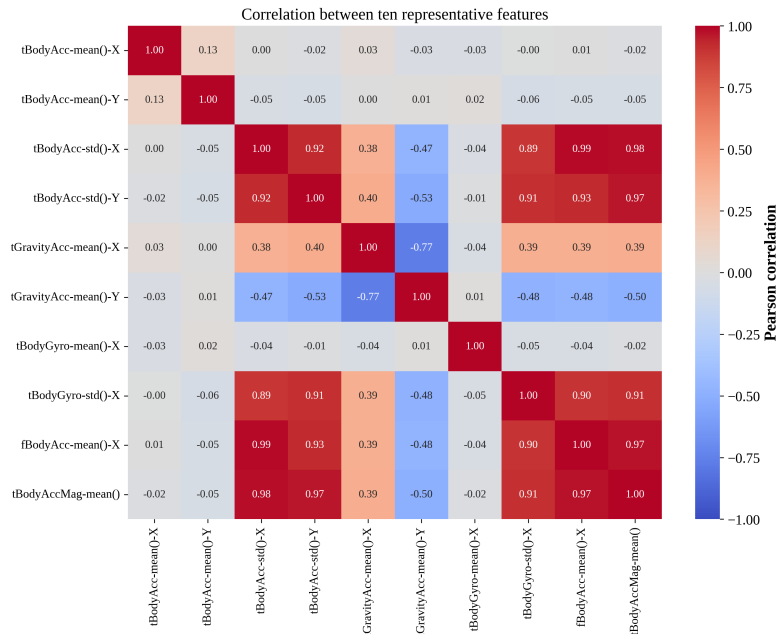


Figure 4: Correlation between ten representative features.

Interpretation: Many HAR features are derived from the same underlying signals (mean, std, energy of the same axis), so strong correlation is expected and confirmed: across all 561 features, 5.15% of the 157,080 feature pairs have $|r| > 0.9$ and 23.08% have $|r| > 0.7$. This high-dimensional, correlated feature space is exactly the setting

where PCA-based visualisation and distance-based clustering both need standardisation, and where a small number of principal components can still capture a large share of the discriminative signal.

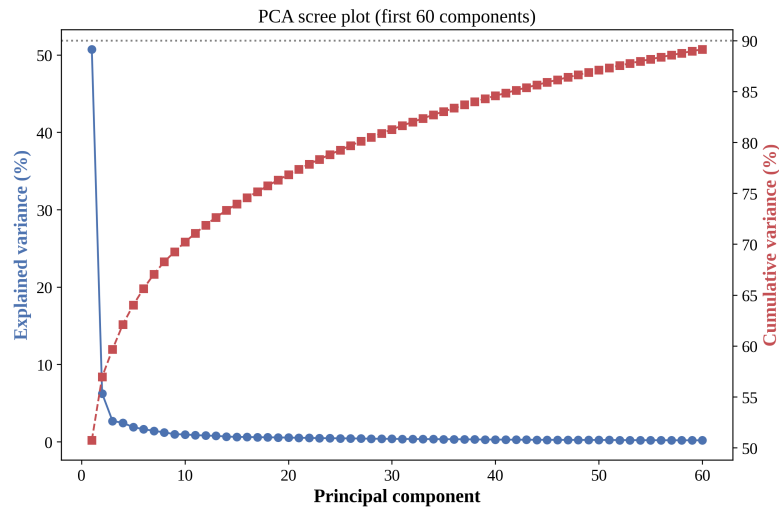


Figure 5: PCA scree plot (individual and cumulative explained variance).

Interpretation: PC1 alone explains 50.7% of the total variance and PC1+PC2 explain 57.0%; 65 components are required for 90% of the variance and 104 for 95%. The first two components therefore capture a genuinely large – but not complete – share of the structure, and are used for every 2-D cluster visualisation in this report; the clustering algorithms themselves are run on the full 561-dimensional standardised data, not on this 2-D projection.

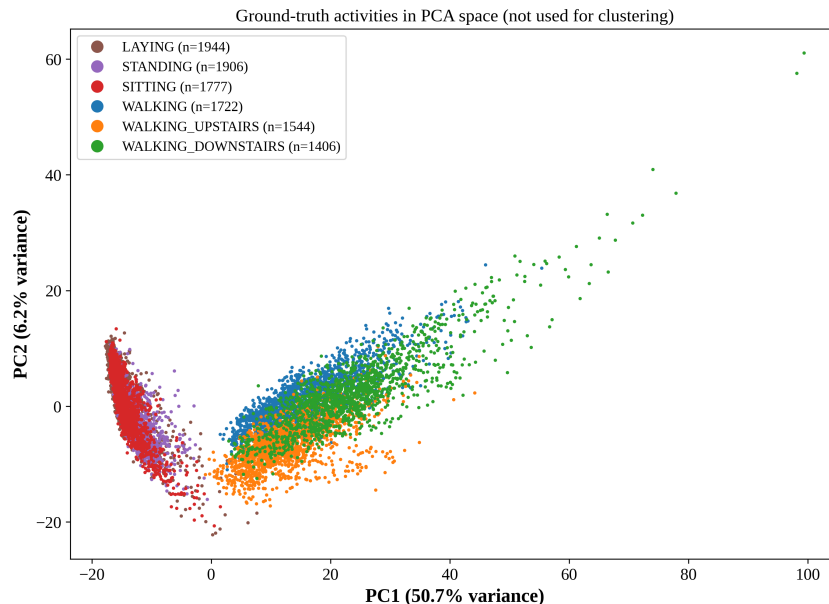


Figure 6: Ground-truth activities in PCA space (labels shown for reference only; never used as clustering input).

Interpretation: PC1 cleanly separates a *static* group (SITTING, STANDING, LAYING, left) from a *dynamic* group (the three WALKING variants, right); within each group

the individual activities overlap substantially in this 2-D projection. This single dominant axis is the main structure that all three unsupervised algorithms are able to recover without seeing the labels, and it explains much of what follows in Sections 6–10.

6 K-Means Clustering (Model A)

K-Means minimises the within-cluster sum of squares, $WCSS = \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$, using Euclidean distance on the standardised 561-D features. It was run for $k = 2, \dots, 8$ with `n_init=10` and `random_state=42`.

6.1 Elbow Method and Silhouette Analysis

Table 3: Table 1 – K-Means elbow/silhouette results (Notebook §7.1)

k	WCSS	Silhouette
2	3,272,857	0.3937
3	2,921,078	0.3155
4	2,781,602	0.1500
5	2,654,790	0.1286
6	2,577,180	0.1099
7	2,519,774	0.0816
8	2,465,379	0.0749

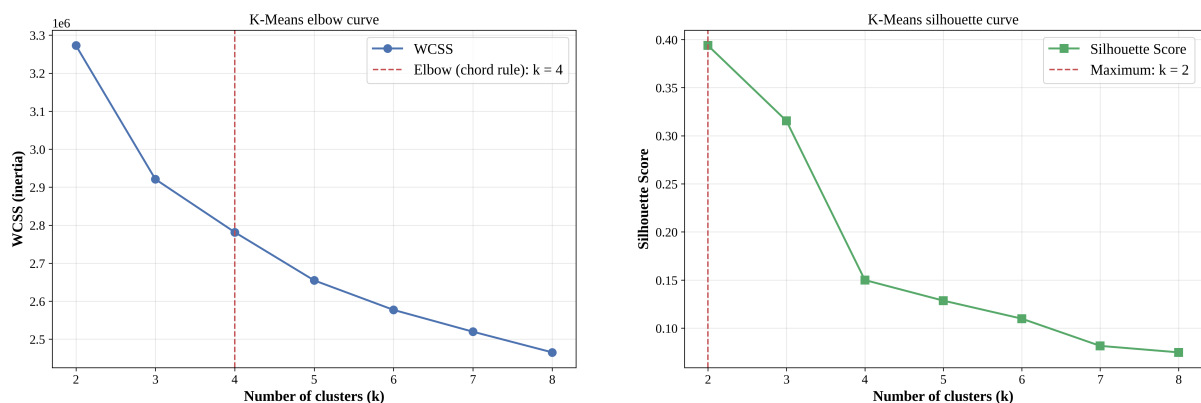


Figure 7: Left: K-Means elbow curve (WCSS vs. k), with the objective elbow (maximum distance-to-chord) at $k = 4$. Right: K-Means silhouette curve, maximised at $k = 2$ (0.394).

Selecting the final k . The number of true activities (6) was deliberately *not* assumed. The final k was chosen using the highest Silhouette Score, an internal criterion computed without any label information, cross-checked against the WCSS elbow located objectively by the maximum-distance-to-chord rule. Silhouette is maximised at $k = 2$ (0.394) and falls monotonically for every larger k ; the WCSS elbow independently falls at $k = 4$. Since the two heuristics disagree, both $k = 2$ (internally selected) and $k = 6$ (the number of true activities, used only as a reference point) are carried forward and compared in Section 11.

6.2 K-Means Cluster Visualisation

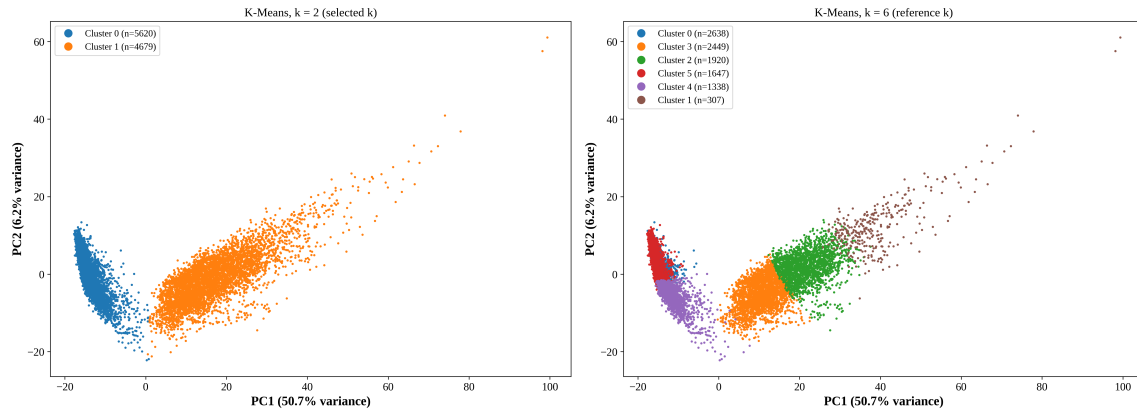


Figure 8: K-Means clusters in PCA space at $k = 2$ (selected) and $k = 6$ (reference).

Interpretation: At $k = 2$, K-Means recovers almost exactly the static-vs-dynamic split seen in the ground-truth plot (Figure 6); this is why it scores so well internally. At $k = 6$, K-Means subdivides both groups further, but the resulting boundaries in PC1–PC2 space do not align as cleanly with any single true activity, since the extra structure that distinguishes, e.g., SITTING from STANDING lies outside the first two principal components.

7 DBSCAN (Model B)

DBSCAN was run on the standardised 561-D data (Euclidean distance). A point is a core point if at least `min_samples` points (itself included) lie within radius `eps`; clusters are density-connected sets of core points plus their border points, and all other points are noise.

7.1 Choosing the Search Range (k-distance plot)

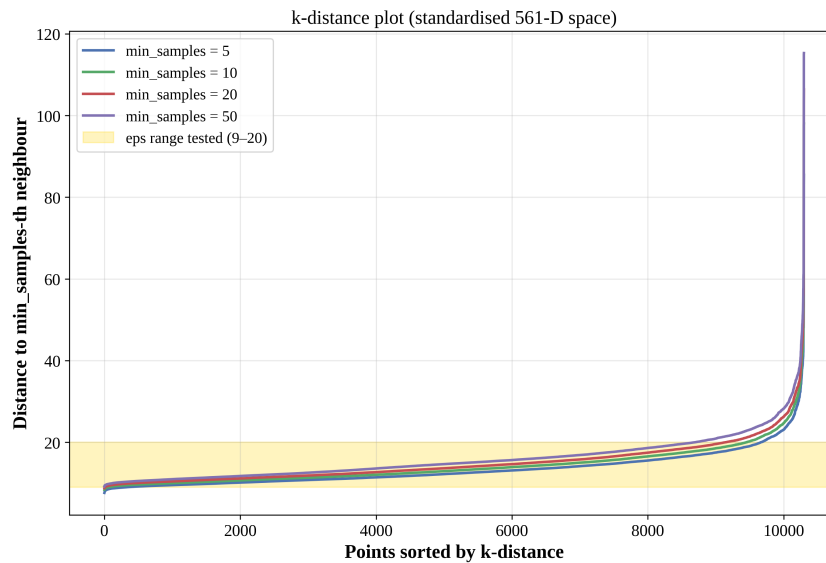


Figure 9: k-distance plot for four values of `min_samples`, with the tested `eps` range (9–20) highlighted.

Interpretation: The distance to the `min_samples`-th nearest neighbour stays low and flat for roughly the first 9,000 points, then rises sharply – the classic “knee” used to bound a sensible `eps` search range. The grid $\text{eps} \in \{9, \dots, 18, 20\} \times \text{min_samples} \in \{5, 10, 20, 50\}$ (44 configurations) was tested; larger `eps` was not explored because the neighbourhood graph becomes near-fully dense and the marginal points added are increasingly non-representative outliers.

7.2 Parameter Tuning

Table 4: Table 3 (excerpt) – Top DBSCAN configurations by Silhouette Score (full 44-row grid in `tables/table3_dbscan_parameter_tuning.csv`)

eps	min_s	clusters	noise	noise%	Sil.	DB	CH	ARI
14	50	2	4126	40.1	0.408	1.057	3343	0.257
12	20	9	5663	55.0	0.374	1.700	189	0.177
13	50	3	5479	53.2	0.364	1.760	226	0.222
13	20	4	4546	44.1	0.352	1.779	1021	0.231
14	20	4	3504	34.0	0.348	1.458	1578	0.275

Across all 44 configurations, 1 found no clusters at all, 13 collapsed into a single cluster, and 30 produced two or more clusters. Noise ranged from 4.2% to 100% of the data depending on the parameters.

Selection rule. A configuration was called *useful* only if it had at least 2 clusters, noise $\leq 20\%$ and no single cluster holding more than 90% of the clustered points; no configuration satisfied this directly, so the rule was relaxed once (noise $\leq 50\%$), giving 8 eligible configurations. Among these, the configuration with the highest Silhouette Score

was selected: **eps** = **14**, **min_samples** = **50** (2 clusters, 4,126 noise points = 40.1%, Silhouette 0.408). This selection was verified two independent ways: (i) against a second DBSCAN run using a completely different (brute Euclidean) code path, and (ii) against a from-scratch re-derivation of the DBSCAN definition using chunked pairwise distances and connected components – both checks matched exactly (noise sets identical, core-point partition ARI = 1.0).

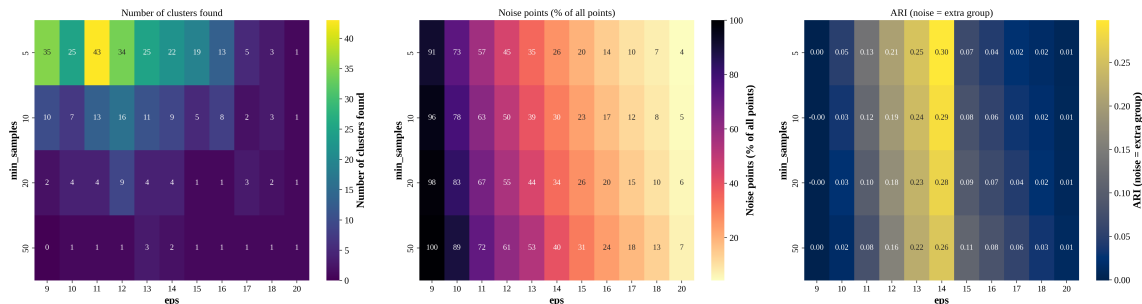


Figure 10: DBSCAN tuning heat-maps: number of clusters, noise percentage, and ARI across the (eps, min_samples) grid.

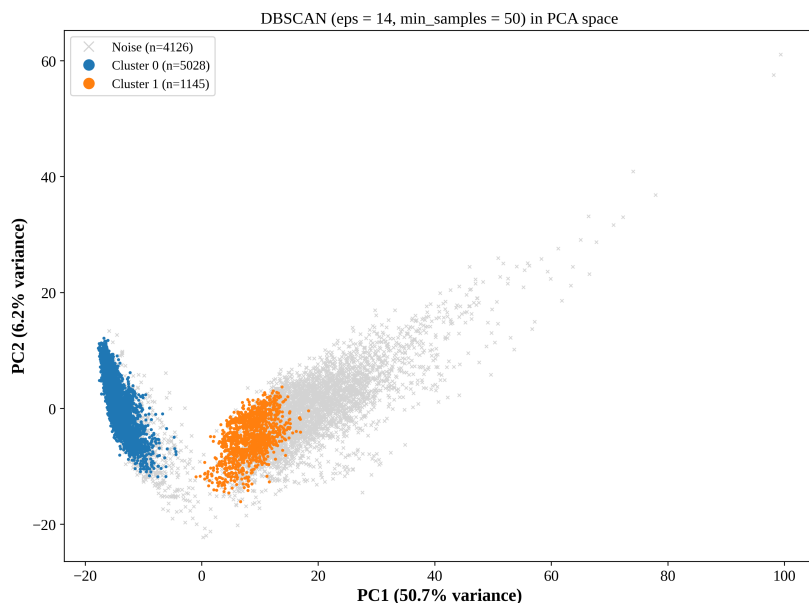


Figure 11: Selected DBSCAN clustering (eps=14, min_samples=50) in PCA space.

Interpretation: DBSCAN recovers the same two dense cores as K-Means at $k = 2$ (compare Figure 8), but treats the diffuse region between the static and dynamic groups – rather than isolated outliers – as noise, marking 40.1% of all points as unclustered. This shows DBSCAN correctly identifies that the transition zone between activity groups is genuinely lower-density, but is not well suited to isolating all six individual activities on this dataset.

8 Hierarchical Agglomerative Clustering (Model C)

Agglomerative clustering was run on the full 10,299-point standardised dataset (no subsampling was needed for the clustering itself: the full condensed Euclidean distance vector has 53.0 million entries, 0.42 GB, which fits in memory) using single, complete, average and Ward linkage. `scipy.cluster.hierarchy.linkage` built each merge tree once; `fcluster` cut each tree at $K = 2, \dots, 8$. Results were cross-checked against `sklearn.cluster.AgglomerativeClustering` (minimum ARI between the two implementations across all checked settings: 1.000, i.e. identical partitions).

8.1 Linkage Comparison

Table 5: Table 4 – Hierarchical linkage comparison at $K = 2$ and $K = 6$ (Notebook §9.2)

Linkage	K	Smallest	Largest	Sil.	DB	CH	ARI	NMI
Single	2	2	10297	0.792	0.275	80.4	0.00003	0.0004
Complete	2	16	10283	0.664	0.760	210.1	0.0002	0.0030
Average	2	3	10296	0.769	0.520	89.8	0.00005	0.0006
Ward	2	4672	5627	0.393	1.072	7859.8	0.333	0.557
Single	6	1	10293	0.560	0.273	28.9	0.0001	0.0013
Complete	6	2	5626	0.379	1.300	1898.4	0.329	0.542
Average	6	1	10141	0.443	1.102	161.1	0.0024	0.0220
Ward	6	151	3208	0.117	2.482	2349.7	0.460	0.601

Interpretation: Single, complete and average linkage all show severe *chaining*: at $K = 2$ the largest cluster already contains 99.8–99.97% of all points, and at $K = 6$ single linkage still places 10,293 of 10,299 points in one cluster. Their high Silhouette scores (up to 0.792) are an artefact of one very large, coherent cluster dominating the average – not evidence of a meaningful 2- or 6-way partition. Ward linkage is the only one that gives balanced clusters at both K values and, at $K = 6$, achieves the best external agreement (ARI 0.460, NMI 0.601) of every configuration tested anywhere in this study.

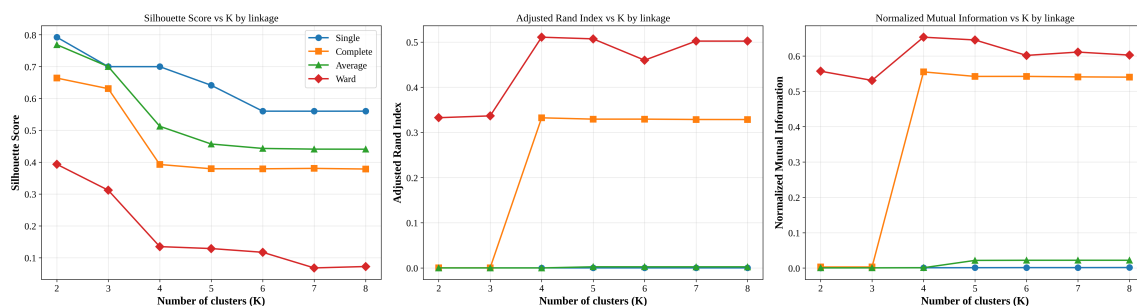


Figure 12: Silhouette, ARI and NMI vs. K for each of the four linkages.

8.2 Dendrograms

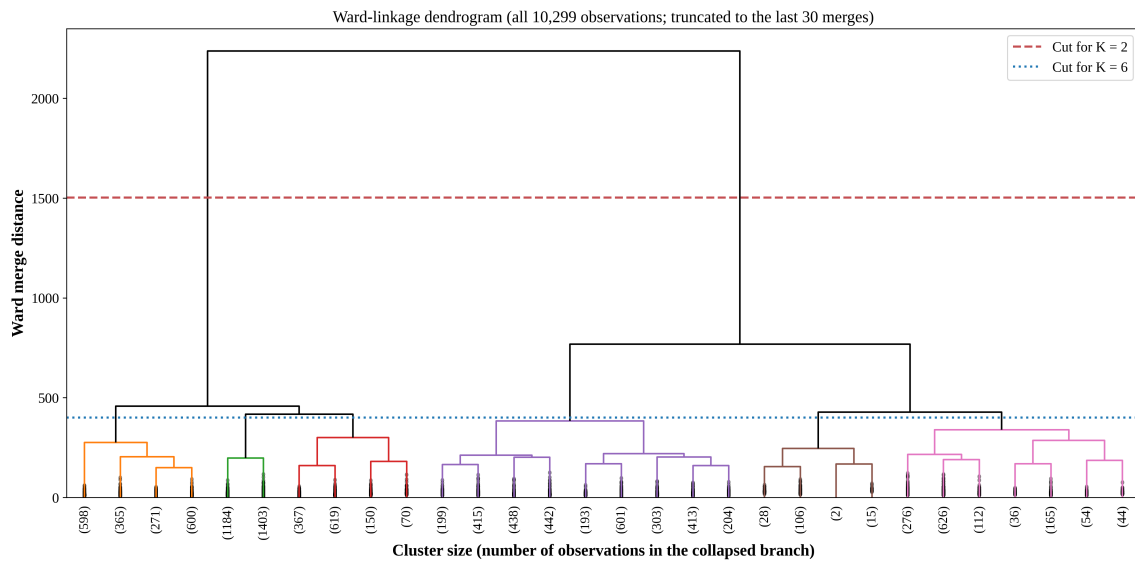


Figure 13: Ward-linkage dendrogram, built from all 10,299 observations and truncated to the last 30 merges for readability (each leaf label is the number of observations it collapses); dashed/dotted lines mark the cuts used for $K = 2$ and $K = 6$.

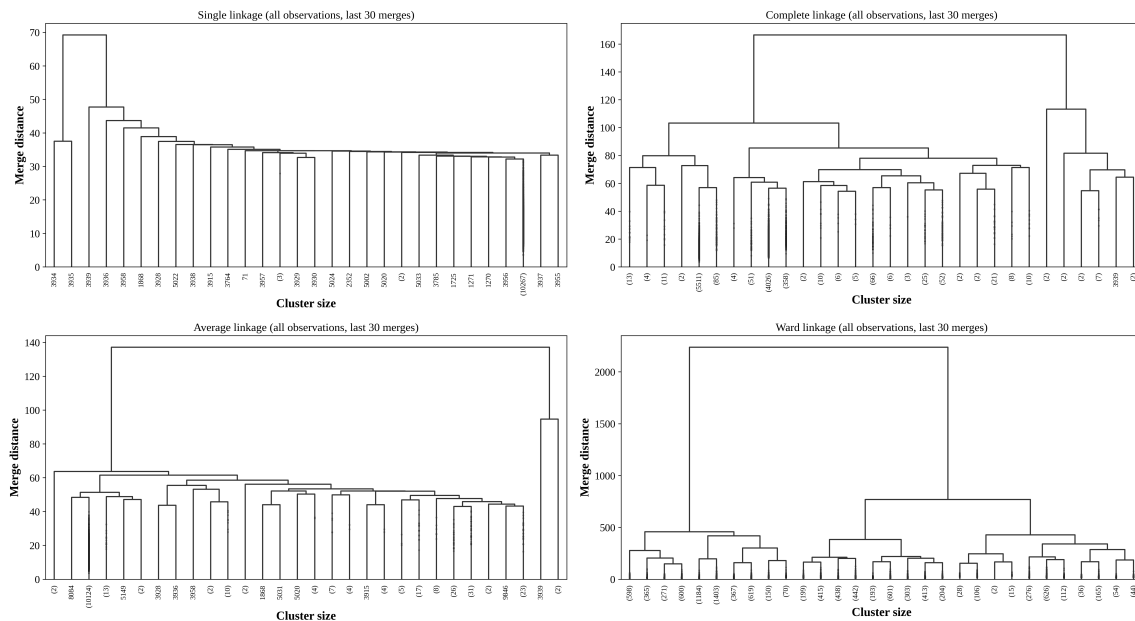


Figure 14: Truncated dendrograms (last 30 merges) for all four linkages, built from all 10,299 observations.

Interpretation: The Ward dendrogram (Figure 13) shows a clean, balanced binary structure typical of variance-minimising merges. The single-linkage dendrogram in Figure 14 visibly shows the chaining pattern discussed above: branches merge into one dominant path rather than forming distinct, well-separated groups.

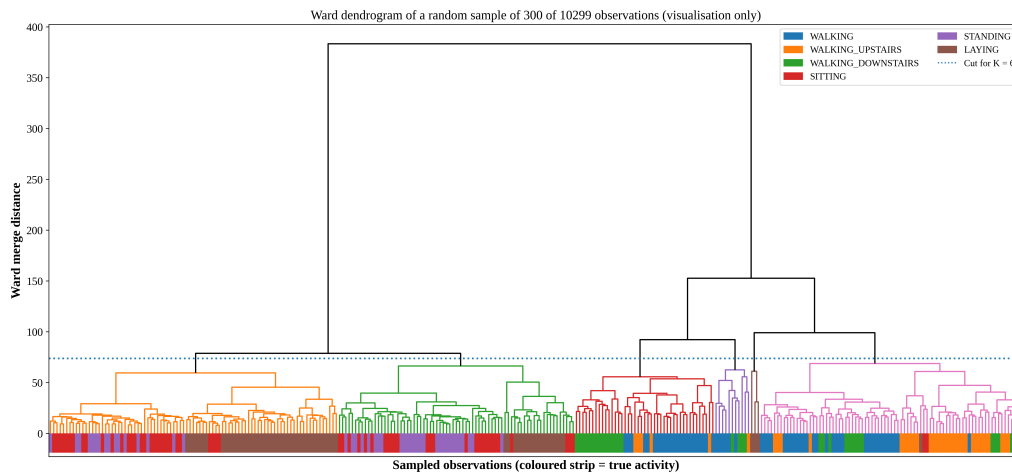


Figure 15: Ward-linkage dendrogram of a random sample of 300 of the 10,299 observations (`random_state=42`), shown for visualisation only; the coloured strip marks each sampled window’s true activity. All reported metrics in this report use the full 10,299-point clustering, not this sample.

Interpretation: The 300-point sample tree lets individual leaves be seen; the activity colour strip shows that nearby leaves tend to share an activity, and that the deepest split broadly separates static (red/purple/brown) from dynamic (blue/orange/green) activities, consistent with the full-data PCA and clustering results.

8.3 HAC Cluster Visualisation

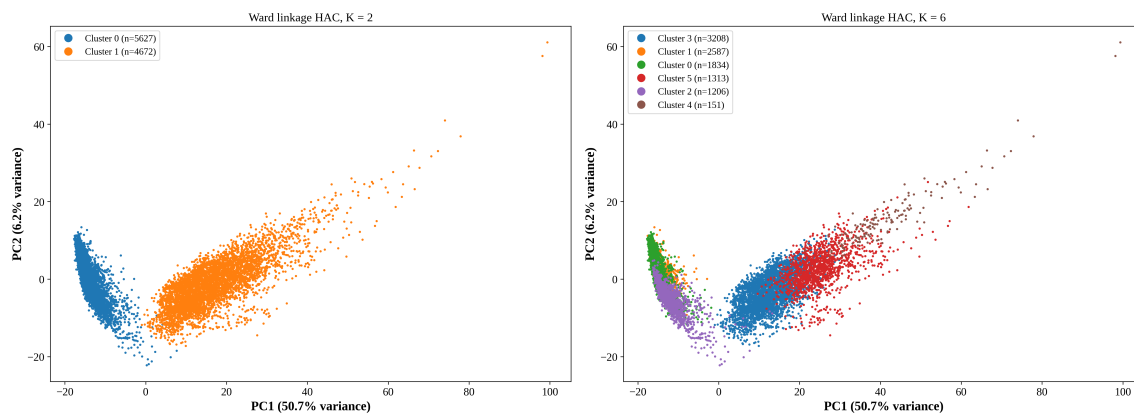


Figure 16: Ward-linkage HAC clusters in PCA space at $K = 2$ (selected) and $K = 6$ (reference).

9 PCA and t-SNE Visualisation

Every cluster visualisation above uses the same 2-D PCA projection (PC1 50.7%, PC2 6.2% of variance). As a non-linear supplementary view, t-SNE (perplexity 30, PCA-initialised, `random_state=42`) was additionally applied to the first 50 principal components; it is used purely for visual inspection, never for any metric.

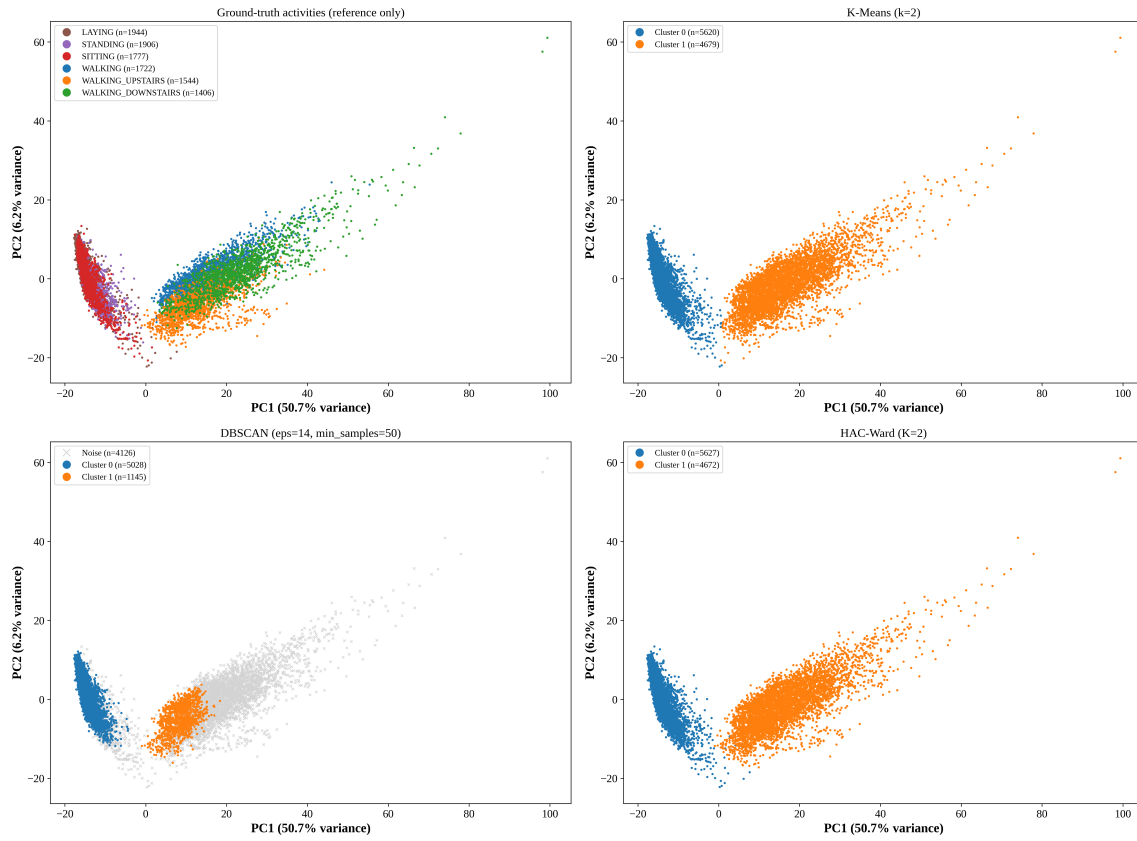


Figure 17: Side-by-side PCA view: ground truth vs. K-Means, DBSCAN, and Ward-HAC (all at their selected $k = 2$ setting where applicable).

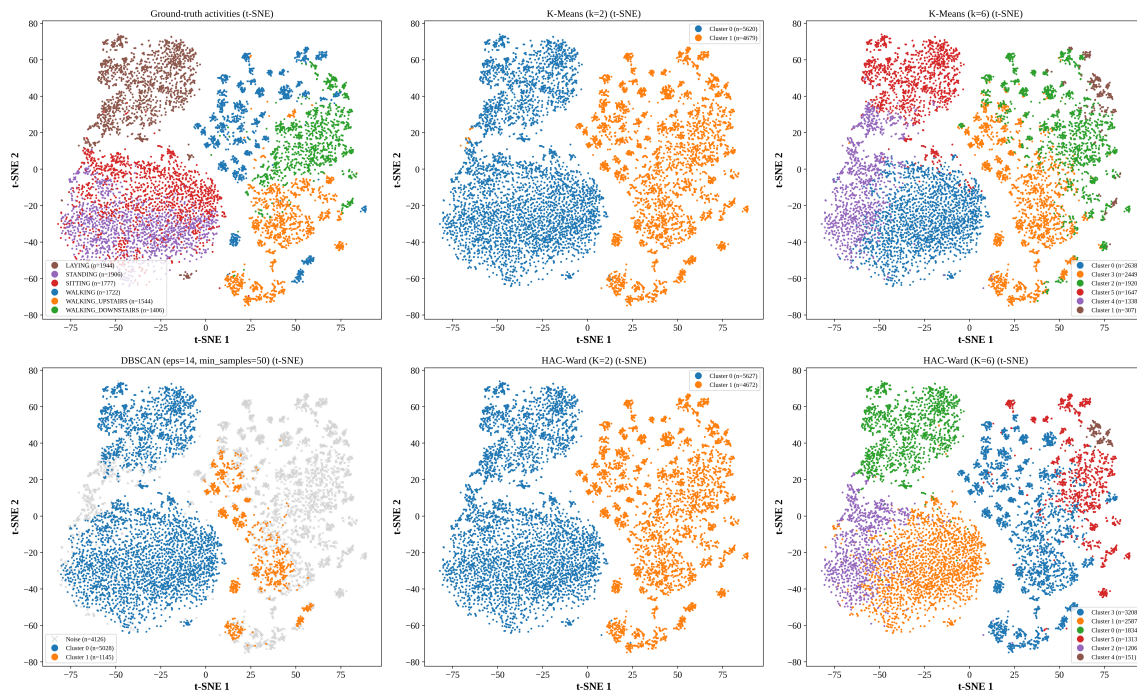


Figure 18: Same comparison using a t-SNE projection.

Interpretation: The t-SNE view confirms the PCA-based conclusions: K-Means and Ward-HAC at $k = 2$ both align closely with the static/dynamic split, and DBSCAN's

noise points concentrate in the transitional region between the two t-SNE lobes, matching the PCA picture.

10 Evaluation Metrics

Internal metrics (Silhouette Score – higher is better; Davies–Bouldin Index – lower is better; Calinski–Harabasz Index – higher is better) were computed on the standardised 561-D data, excluding DBSCAN noise points (for which no cluster is defined). External metrics (Adjusted Rand Index and Normalized Mutual Information against the six true activities, both higher is better) were computed with DBSCAN noise treated as its own group (so that discarding points is not rewarded); the values restricted to only the clustered (non-noise) points are also reported where relevant.

11 Results Comparison

Table 6: Table 2 – Final clustering comparison (Notebook §11)

Algorithm	Parameters	Clusters	Noise	Sil.	DB	CH	ARI	NMI
K-Means	$k=2$	2	0	0.394	1.071	7880.8	0.330	0.545
K-Means	$k=6$	6	0	0.110	2.384	2556.5	0.420	0.559
DBSCAN	$\text{eps}=14, \text{min_s}=50$	2	4126	0.408	1.057	3343.4	0.257	0.405
HAC (Ward)	$K=2$	2	0	0.393	1.072	7859.8	0.333	0.557
HAC (Ward)	$K=6$	6	0	0.117	2.482	2349.7	0.460	0.601

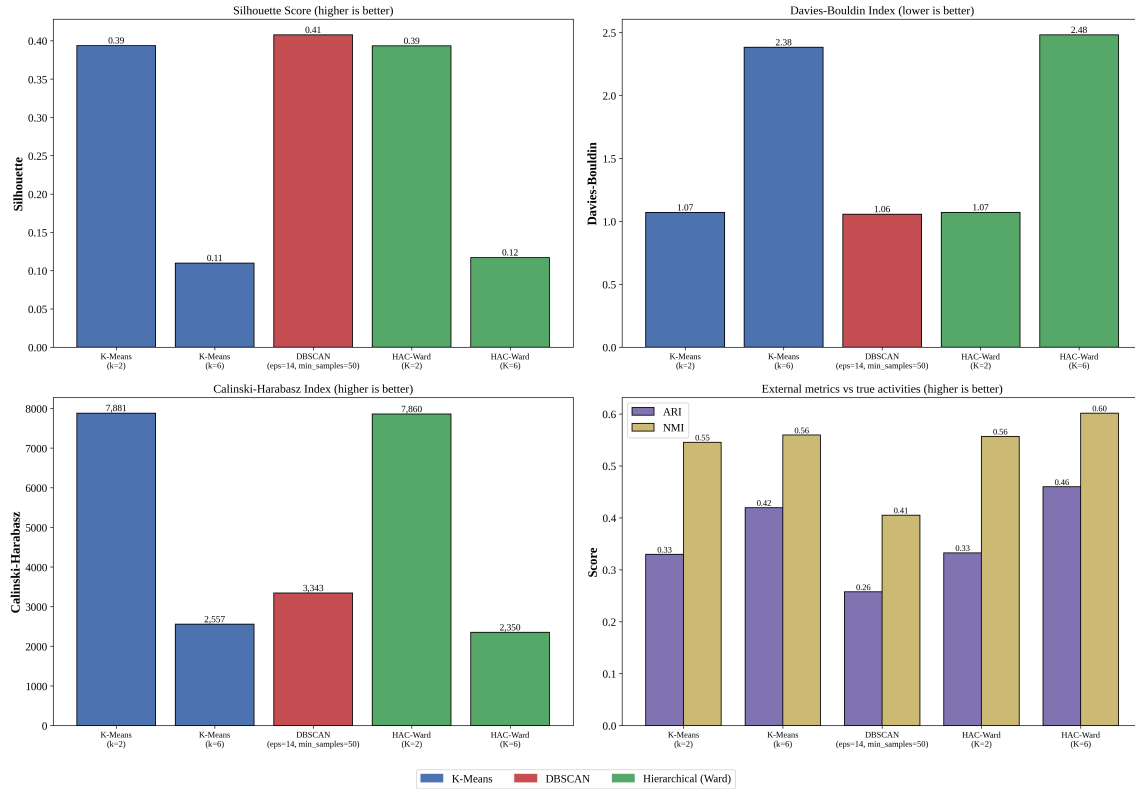


Figure 19: All five internal/external metrics compared across the five final models.

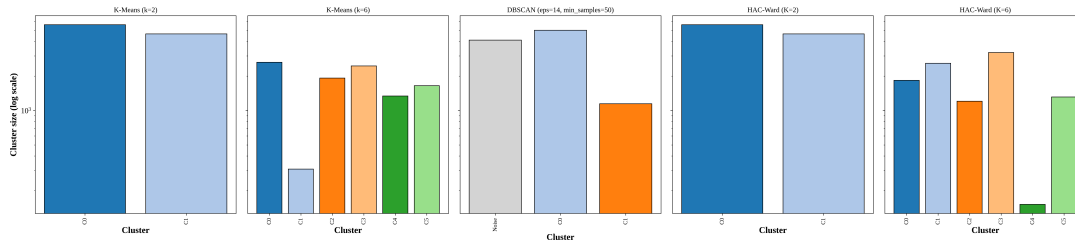


Figure 20: Table 5 – Cluster size distribution (log scale) for each final model.

Interpretation: By every internal metric, K-Means and Ward-HAC at $k = 2$ are essentially tied for best, with DBSCAN's selected configuration scoring marginally higher on Silhouette only because 40% of the data was excluded from the calculation. By external agreement with the six true activities, Ward-HAC at $K = 6$ is clearly the strongest (ARI 0.460, NMI 0.601), ahead of K-Means at $k = 6$; DBSCAN is weakest externally among the five final models.

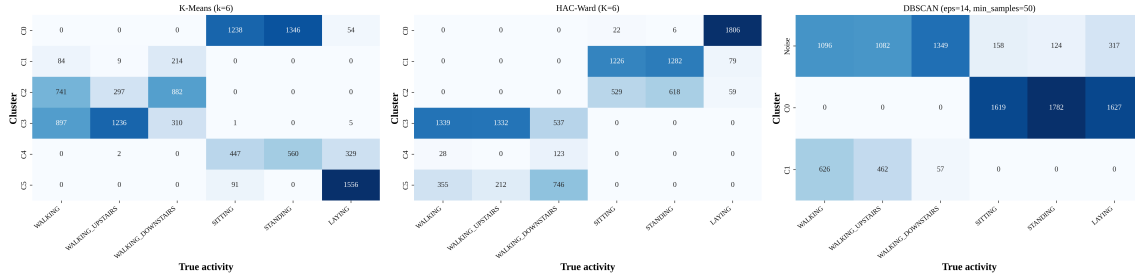


Figure 21: Cluster-vs-activity contingency tables (rows = cluster, columns = true activity) for K-Means ($k=6$), Ward-HAC ($K=6$) and DBSCAN.

Interpretation: Both $K/k = 6$ solutions produce clusters that are each dominated by one or two true activities rather than a uniform mixture, confirming that meaningful (if imperfect) activity-level structure is being recovered; the confusion is concentrated between similar activities (e.g. the three WALKING variants, or SITTING vs. STANDING), consistent with their proximity in the PCA/t-SNE plots.

12 Observation Questions

1. Which algorithm produced the most meaningful clusters? Why?

This depends on the criterion. By internal criteria, K-Means and Ward-HAC at $k = 2$ are tied for best (Silhouette ≈ 0.39 , Davies–Bouldin ≈ 1.07) and represent a genuine, high-density boundary in the data – the static-vs-dynamic split visible in Figure 6. By external agreement with the six true activities, Ward-linkage HAC at $K = 6$ is best (ARI 0.460, NMI 0.601), ahead of K-Means at $k = 6$ (ARI 0.420) and well ahead of DBSCAN (ARI 0.257). Since the task is unsupervised and the six labels are for validation rather than model selection, both answers are legitimate; the disagreement between the internally-best and externally-best solutions is itself a central finding of this experiment.

2. How sensitive was K-Means to the choice of k ?

Very sensitive, and monotonically so on the internal metric: Silhouette falls from 0.394 at $k = 2$ to 0.075 at $k = 8$ with no local maximum in between (Table 3), while WCSS shows a single visible bend at $k = 4$ – the two standard heuristics disagree on the “right” k . External agreement moves differently: ARI dips slightly through $k = 5$, then rises to 0.420 at $k = 6$ and peaks at 0.433 at $k = 7$, i.e. beyond $k = 5$ more clusters improve the match to ground truth even as every internal score keeps getting worse – clear evidence that K-Means, left to its own objective, is biased toward fewer, well-separated clusters and will not discover $k = 6$ unaided.

3. Did DBSCAN detect noise or small clusters effectively?

Partially. The 44-configuration grid search shows DBSCAN is very sensitive to its parameters on this dataset: 13 configurations collapsed to a single cluster and one found none at all, but in the useful middle of the grid DBSCAN does find genuinely small clusters (e.g. $\text{eps}=13$, $\text{min_samples}=5$ gives 25 clusters, the smallest holding only 5 points), something K-Means cannot represent. However, at the internally-selected configuration, the 40.1% of points marked as noise are concentrated in the diffuse transition region between the static and dynamic activity groups (Figure 11) rather than behaving like scattered outliers, so DBSCAN is effective at identifying that this region is genuinely lower-density, but not at treating it as anything other than unclusterable, and no configuration in the grid found six low-noise clusters matching the six true activities.

4. How does linkage choice (single/complete/ward) affect hierarchical clustering?

Substantially. Single, complete and average linkage all exhibit severe chaining on this high-dimensional data – at $K = 2$ their largest cluster holds 99.8–99.97% of all 10,299 points – and their apparently high Silhouette scores are an artefact of that one dominant cluster rather than evidence of meaningful structure. Ward linkage is the only one of the four that gives balanced partitions at both $K = 2$ and $K = 6$ and is also the only linkage whose external metrics beat the corresponding K-Means solution at $K = 6$, consistent with Ward’s variance-minimisation objective being much closer to K-Means’ own objective than single/complete/average linkage.

5. Which internal metric best matched visual intuition?

Davies–Bouldin and Calinski–Harabasz matched the (label-based) sense of cluster quality far more consistently than Silhouette did. Over the 35 K-Means and HAC partitions evaluated (Table 7), Davies–Bouldin correlates with ARI at $\rho = 0.879$ and Calinski–Harabasz at $\rho = 0.791$, while Silhouette correlates at $\rho = -0.900$ – i.e. Silhouette moves in essentially the opposite direction from the external, ground-truth-based metrics. This matches the chaining discussion in Q4: Silhouette rewards the visually degenerate single/complete/average solutions (one huge blob plus a handful of points), while Davies–Bouldin and Calinski–Harabasz, through their explicit use of cluster centroids and compactness, penalise that same structure and agree better with what the PCA/t-SNE plots show by eye.

Table 7: Table 6 – Internal-metric agreement with external metrics (Spearman ρ , Notebook §12)

Partitions	Internal metric	ρ vs. ARI	ρ vs. NMI
K-Means only (7)	Silhouette	−0.643	−0.500
K-Means only (7)	Davies–Bouldin	0.536	0.321
K-Means only (7)	Calinski–Harabasz	−0.643	−0.500
HAC only (28)	Silhouette	−0.917	−0.936
HAC only (28)	Davies–Bouldin	0.882	0.882
HAC only (28)	Calinski–Harabasz	0.873	0.838
K-Means+HAC (35)	Silhouette	−0.900	−0.885
K-Means+HAC (35)	Davies–Bouldin	0.879	0.847
K-Means+HAC (35)	Calinski–Harabasz	0.791	0.766

Note: a negative ρ for Davies–Bouldin (lower is better) would indicate agreement with the external metrics; the positive values above indicate agreement in the expected direction. For Silhouette and Calinski–Harabasz (higher is better) a positive ρ indicates agreement, so the negative values above indicate disagreement.

13 Comparative Analysis

Table 6 and Figure 19 summarise the trade-off explored throughout this report: **K-Means and Ward-linkage HAC at $k = 2$** give the internally cleanest partition, recovering the single dominant static-vs-dynamic axis of variation with high compactness and separation but modest agreement with the six true activities ($\text{ARI} \approx 0.33$). **Ward-linkage HAC at $K = 6$** sacrifices internal compactness (Silhouette drops to 0.117) in

exchange for the best available match to the true activity structure (ARI 0.460), making it the strongest overall result when validated against ground truth. **DBSCAN** occupies a different niche: rather than partitioning every point, it identifies which points sit in an unambiguous, dense region of feature space (the two activity groups) versus which sit in the ambiguous transition between them, at the cost of not clustering 40% of the data and of the weakest external agreement of the three algorithms. **K-Means at $k = 6$** sits between K-Means at $k = 2$ and Ward-HAC at $K = 6$ on every metric, confirming that Ward’s variance-minimising, hierarchy-aware merges give it a genuine edge over K-Means specifically at the activity-level granularity, even though the two algorithms are near-identical at the coarser $k = 2$ granularity.

14 Conclusion

This experiment applied K-Means, DBSCAN and Agglomerative Clustering to the 561-feature, 10,299-observation UCI HAR dataset without ever exposing the six activity labels to any clustering algorithm. Standardisation was necessary (raw feature standard deviations varied $20.2\times$) and effective. PCA revealed one dominant axis of variation separating static from dynamic activities (50.7% of total variance on PC1 alone), which every algorithm was able to recover to some degree.

K-Means, scored purely on its own Silhouette objective, selected $k = 2$ and could not be induced to prefer $k = 6$ by any internal criterion, even though $k = 6$ (via Ward-linkage HAC) gave the best match to the ground truth of any configuration tested. DBSCAN, tuned over a 44-configuration grid and independently verified, could not find six well-separated, low-noise clusters anywhere in that grid; at its internally best setting it recovered the same two-group structure as K-Means at $k = 2$ while marking 40.1% of the data as noise concentrated in the transition region between the groups. Hierarchical clustering showed a strong dependence on linkage choice: single, complete and average linkage all suffered severe chaining, while Ward linkage alone produced balanced, externally meaningful partitions at every K tested. Finally, across 35 evaluated partitions, Davies–Bouldin and Calinski–Harabasz agreed with the external, label-based metrics far more consistently than Silhouette did, so Silhouette Score alone – although a valid and correctly computed internal measure – was an unreliable guide to which clustering best matched the known activities on this dataset, and should be interpreted together with the other internal metrics and a visual check rather than in isolation.

Overall, no single algorithm was uniformly best: the choice depends on whether the objective is internal compactness/separation (K-Means/Ward-HAC at $k = 2$), best match to known activity labels (Ward-HAC at $K = 6$), or explicit identification of an ambiguous transition region in the data (DBSCAN). This is consistent with the underlying structure of the HAR dataset: activities are distinguished mainly by one dominant, well-separated static/dynamic mode, with finer distinctions between individual activities corresponding to smaller, less-separated sub-structure that purely unsupervised methods – lacking access to the labels – recover only partially.

15 References

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